A Calibration and Selective-Prediction Benchmark for Perturbation-Effect Uncertainty: Distribution-Free Methods Hold, Parametric Priors Miscover, and a Genetics Taxonomy Does Not Help

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Abstract

We introduce a rigorous, donor-blocked calibration benchmark for uncertainty quantification (UQ) of single-cell perturbation effects, built on a CD4$^+$ T-cell Perturb-seq assay and an independent K562 Perturb-seq assay. The benchmark holds the point predictor fixed across all UQ methods so that any difference in marginal coverage is attributable to the calibration layer alone, and it evaluates methods on a leave-one-donor-out (LODO) split of $58{,}558$ test rows spanning $2{,}591$ perturbations and four donors. Our headline result is an *honest tie*: six distribution-free methods (jackknife+, Mondrian, weighted and split conformal, naive-global quantile, and quantile regression) are statistically indistinguishable in mean absolute marginal coverage error, and the best conformal method does not separate from the strongest non-conformal distribution-free baseline (paired cluster-bootstrap Δ confidence interval includes zero). The benchmark does, however, resolve a clear and significant contrast: distribution-free calibration decisively outperforms a parametric Gaussian prior, which over-covers by roughly 6.7 percentage points at the 80% level under donor shift. We further contribute (i) a cross-assay transfer protocol showing that frozen quantiles fail ($4.8/7.6/17.1\%$ empirical coverage at nominal $80/90/95\%$) but that re-fitting the identical method on the target assay restores near-nominal coverage across six calibration fractions and 500 resplits; (ii) a distribution-free risk-control certificate that is valid per-assay yet honestly vacuous at $n{=}4$ exchangeable donors; and (iii) a pre-registered negative showing that conditioning conformal prediction on an external eQTL taxonomy does not sharpen intervals. All results were hash-frozen and pre-registered before scoring, an independent audit re-derived the benchmark surface from the raw 16.87 GB data with zero discrepancy, and we openly report the self-retraction of two earlier surrogate "wins." This is a benchmark and methods contribution; no biological target is validated in scope.

1 Introduction

Machine-learning models for single-cell perturbation prediction are proliferating, but a growing body of evidence suggests that headline accuracy numbers in this field are frequently inflated. Recent audits report that deep-learning perturbation-effect predictors do not yet outperform simple linear baselines [Ahlmann-Eltze et al., 2025], and systematic benchmarks find that mean baselines are competitive with foundation models on standard scores [Wu et al., 2024]. Much of this fragility traces to a single methodological gap: point-prediction metrics reward memorization of population structure and say nothing about whether a model *knows what it does not know*. A prediction of a perturbation effect that is confidently wrong is more dangerous than one that is honestly uncertain, yet the community lacks a rigorous, shared protocol for measuring calibrated uncertainty under the distribution shifts that matter for this data, namely shifts across biological donors and across assays.

We address this gap with a calibration and selective-prediction benchmark rather than a new predictor. Our motivating application is CD4$^+$ T-cell Perturb-seq relevant to autoimmune biology, but we stress at the outset that this is a measurement contribution: we ask whether the uncertainty around a perturbation-effect estimate is correct and useful, not whether any specific gene is a validated disease target. The disease context supplies the data and the donor structure; it supplies no biological claim.

The benchmark is designed around one controlling principle. Because different UQ methods can differ both in their point predictions and in their calibration, comparisons in the literature often conflate the two. We eliminate that confound by fixing an identical base point predictor (a histogram gradient-boosted regressor) across every method, so that the only quantity that varies is the interval-construction layer. On top of this fixed surface we evaluate nine UQ methods under a donor-blocked leave-one-donor-out (LODO) split, which is the assumption-faithful granularity for the donor shift that practitioners actually face.

Our contributions are as follows.

- **A powered, point-prediction-controlled head-to-head of nine UQ methods** on $58{,}558$ LODO test rows across four donors, with paired cluster-bootstrap significance testing over $2{,}591$ perturbations. The result is an *honest tie* among six distribution-free methods, and a significant, reproducible loss for parametric priors.
- **A cross-assay transfer protocol** demonstrating that frozen calibration quantiles fail on an independent assay but that the underlying method transfers once recalibrated, characterized across six calibration fractions and 500 resplits.
- **A distribution-free risk-control analysis** (RCPS / Learn-then-Test) that yields a valid non-vacuous certificate per assay and honestly reports vacuity at $n{=}4$ donors, quantifying the price of exchangeability rather than hiding it.
- **A pre-registered negative result** showing that conditioning conformal prediction on an external cis-eQTL taxonomy does not produce sharper intervals at matched coverage.
- **An integrity protocol**: hash-frozen pre-registration before scoring, an independent audit that re-derives the benchmark surface from the raw data, and an explicit self-retraction of two earlier surrogate results.

2 Benchmark Design

2.1 Data and target

The primary benchmark surface is derived from a CD4$^+$ T-cell Perturb-seq assay in primary human T cells [Schmidt et al., 2022]. For each perturbation we define the effect target as $y = \log(1 + \|z\|_2)$, where z is the vector of differential-expression z-scores over a shared readout gene panel with the on-target gene excluded, and the z-scores are Welch two-sample statistics computed against negative-control cells. The independent transfer assay is the Norman et al. K562 genome-scale genetic-interaction Perturb-seq dataset [Norman et al., 2019] (GEO accession GSE133344), scored with an identical target definition over a shared panel of $9{,}780$ genes (95.1% of the primary panel).

2.2 The donor-blocked LODO surface

The benchmark evaluates on a leave-one-donor-out split with four donor folds. Each fold holds out all cells from one donor for testing and calibrates on the remaining three, so that no donor appears in both calibration and test. The pooled test set comprises $58{,}558$ rows spanning $2{,}591$ unique perturbations, with per-fold test sizes of $14{,}640$, $14{,}639$, $14{,}639$, and $14{,}640$ rows. This design targets the shift that matters in practice: a model calibrated on some donors must produce trustworthy uncertainty on a new donor.

2.3 The point-prediction control

The central design choice is that an identical base point predictor and identical point predictions are held fixed across all nine UQ methods. Consequently, any difference in marginal coverage or interval width reflects the calibration layer alone and not a difference in the underlying regression. We verified that the committed split-conformal method reproduces its receipted empirical coverage exactly ($0.7932/0.8955/0.9486$ at $80/90/95\%$) on this surface, confirming that the benchmark harness faithfully re-scores the committed pipeline rather than a surrogate.

2.4 Methods compared

We evaluate nine UQ methods that fall into three families. The *distribution-free conformal* family comprises split conformal, Mondrian conformal conditioned on culture condition [Vovk et al., 2005], weighted conformal [Tibshirani et al., 2019], and the jackknife+ [Barber et al., 2021]. The *distribution-free adaptive* family comprises quantile

Table 1: Head-to-head of nine UQ methods on the donor-blocked LODO surface ($58{,}558$ test rows, four donor folds). Point predictions are held identical across all methods. Coverage error is the mean absolute marginal coverage error across the $80/90/95\%$ levels. Width is the mean interval width at 90%. Lower is better for both. The six distribution-free methods (top block) are statistically indistinguishable in coverage error; the two parametric methods (bottom block) miscover.

Method	Family	Coverage error ↓	Width @90% ↓
Jackknife+	distribution-free	**0.0018**	0.4605
Mondrian conformal	distribution-free	0.0037	0.4581
Weighted conformal	distribution-free	0.0037	0.4589
Naive-global quantile	distribution-free	0.0040	0.4581
Split conformal	distribution-free	0.0042	0.4578
Quantile regression	distribution-free	0.0049	0.4869
CQR	distribution-free	0.0054	0.4862
Parametric Gaussian	parametric	0.0339	0.5328
Ensemble-variance Gaussian	parametric	0.7887	0.0362

regression, conformalized quantile regression (CQR) [Romano et al., 2019], and a naive-global quantile baseline. The *parametric* family comprises a parametric Gaussian interval and a naive Gaussian ensemble-variance interval. All methods target marginal coverage at the 80, 90, and 95% levels.

2.5 Evaluation metrics and significance testing

The primary metric is the mean absolute marginal coverage error across the three levels, defined as the mean of $|\hat{c}_\ell - \ell|$ for $\ell \in \{0.80, 0.90, 0.95\}$, where $\hat{c}_\ell$ is the empirical coverage. We report interval width as an efficiency metric and area under the risk-coverage curve (AURC) for selective prediction. Significance is assessed with a paired cluster bootstrap (2,000 replicates) that resamples the $2{,}591$ perturbations as clusters, which respects the dependence induced by shared perturbations across rows and yields confidence intervals on pairwise differences between methods.

3 Results

3.1 The core result: an honest tie among distribution-free methods

Table 1 reports mean absolute coverage error and mean interval width at 90% for all nine methods. Six distribution-free methods form a tight cluster in coverage error, from jackknife+ at 0.0018 through Mondrian and weighted conformal at 0.0037, naive-global at 0.0040, split conformal at 0.0042, quantile regression at 0.0049, and CQR at 0.0054. The paired cluster bootstrap confirms that this cluster is statistically indistinguishable: the comparison between the best committed conformal method (Mondrian) and the strongest non-conformal distribution-free baseline (quantile regression) gives a Δ of -0.00074 with a 95% confidence interval of $[-0.0034, +0.0023]$, which includes zero. Similarly, jackknife+ versus split ($[-0.0028, +0.0025]$), Mondrian versus split ($[-0.0014, +0.0007]$), and CQR versus split ($[-0.0021, +0.0031]$) all straddle zero.

This is the honest headline. On this surface, conformal prediction does not beat every prior-art distribution-free method; quantile regression, itself a distribution-free adaptive method, ties it. We report the tie because it is the truth of the measurement, and because a tie among principled distribution-free methods is itself a useful result for practitioners choosing a calibration layer.

3.2 The significant contrast: parametric priors miscover under shift

The benchmark does resolve a clear and significant difference: the distribution-free family decisively outperforms the parametric family. The parametric Gaussian interval over-covers, reaching 86.67% empirical coverage at the nominal 80% level, an over-coverage of 6.7 percentage points, with a mean absolute coverage error of 0.0339, nearly an order of magnitude worse than the distribution-free cluster. The paired cluster bootstrap places the split-versus-parametric difference at -0.0287 with a 95% confidence interval of $[-0.0376, -0.0152]$, which excludes zero; the Mondrian-versus-parametric difference is -0.0292 with interval $[-0.0373, -0.0159]$, also excluding zero. The naive Gaussian ensemble-variance interval collapses entirely, achieving roughly 8% coverage at the 90% level (mean absolute coverage error 0.7887), because its variance estimate badly underestimates the residual spread.

Two secondary observations sharpen the picture. First, adaptive-width methods buy no sharpness here: at approximately equal coverage, the CQR 90% interval is significantly *wider* than split conformal by $+0.028$ $\log_2$FC units (interval $[+0.0278, +0.0291]$), so CQR does not Pareto-dominate the simpler incumbent, a consequence of the residual distribution being close to homoscedastic in these leakage-safe features. Second, on selective prediction all methods sit in a narrow AURC band of roughly 0.112 to 0.120, far above the oracle lower bound of 0.048 and close to the no-skill random line of 0.118, indicating that uncertainty-based trust ordering is only weakly informative on this surface and does not separate the methods.

3.3 Cross-assay transfer: frozen quantiles fail, recalibration fixes

Calibration guarantees are exchangeability-bound, so a natural benchmark question is whether quantiles calibrated on one assay transfer to another. We applied the frozen $CD4^+$ conformal quantiles unchanged to the independent K562 assay. Coverage collapsed to 4.8/7.6/17.1% at nominal 80/90/95%: a decisive failure. A diagnostic decomposition shows the failure is both location and scale, since the K562 residual spread is $2.6\times$ wider than the $CD4^+$ calibration spread, so even oracle mean-recentering leaves coverage at only 0.41/0.49/0.68. We report this as a negative rather than tuning it away; it is the expected behavior of split conformal under distribution shift.

Crucially, the *method* transfers even though the specific quantiles do not. Re-fitting the identical split-conformal procedure on the K562 assay's own calibration split restores near-nominal coverage on a held-out K562 test split: a primary 50/50 split gives 0.849/0.887/0.943, and a robustness sweep over 500 random resplits gives mean coverage of 0.755/0.864/0.927, all three within the pre-registered ± 0.05 band of nominal. A frozen-base sensitivity variant that recalibrates only the quantiles (keeping the $CD4^+$ base predictor fixed) is even tighter at 0.792/0.886/0.943, showing that the assay-bound quantity is the quantile and not the base predictor. Figure-level robustness holds at every calibration fraction from 0.2 to 0.7: at each fraction the 95% resplit distribution of coverage brackets the nominal level. This yields a portable, cross-dataset calibration protocol with an honest failure mode and its fix.

3.4 Distribution-free risk control, honestly scoped

We ask whether a finite-sample distribution-free certificate on miscoverage can be issued using Risk-Controlling Prediction Sets (RCPS) [Bates et al., 2021] and Learn-then-Test [Angelopoulos et al., 2021]. The certificate is exactly as strong as its exchangeability precondition. Under donor-blocked LODO the assumption-faithful exchangeable unit is the donor, of which there are only four. At n=4 donors the distribution-free upper confidence bound on miscoverage is 0.74/0.64/0.59 (Hoeffding) against targets of 0.20/0.10/0.05: the certificate is honestly *vacuous*. A row-level version that appears tight (n=58,558) rests on row exchangeability across the donor boundary, which is precisely the assumption the benchmark flags as violated, so we report it only as a false-precision contrast, not a guarantee.

In the per-assay recalibrated regime, however, exchangeability holds by construction under a random calibration/test split of the 105 K562 single-gene perturbations, and the certificate becomes valid and non-vacuous. RCPS certifies a half-width $\hat{\lambda} = 0.527$ at the 90% level (upper confidence bound $0.099 \leq 0.10$) with realized held-out coverage of 0.943; certificates at 80/90/95% all hold with realized test coverage 0.887/0.943/0.943. The contribution here is to make the price of exchangeability explicit and quantified: risk control is deployable per assay but cannot certify the donor-shift setting at n=4, and we say so rather than overclaiming.

3.5 A pre-registered negative: a genetics taxonomy does not help

A plausible idea for sharpening perturbation intervals is to condition the conformal calibration on an external, orthogonal genetics channel, partitioning perturbations by a cis-eQTL tier and running Mondrian conformal per tier. We pre-registered this hypothesis and tested it on the 612 genetics-annotated LODO test rows (tier sizes 360/168/84). The genetics-conditioned Mondrian method does not sharpen intervals relative to plain split conformal; if anything it makes them *wider*. The paired-bootstrap difference in mean interval width (genetics-Mondrian minus split) is $+0.020$ at 80% (CI $[+0.001, +0.034]$, excludes zero), $+0.022$ at 90% (CI $[-0.016, +0.057]$, includes zero), and $+0.219$ at 95% (CI $[+0.032, +0.438]$, excludes zero). At two of the three levels the external genetics taxonomy is thus *significantly worse* than plain split conformal, and at no level is it better. Coverage was held (tier coverages of $0.897/0.887/0.845$ at the 90% level, all with confidence intervals overlapping nominal). We report this as a first-class negative result: it saves other groups from pursuing an intuitive but empirically empty (indeed counterproductive) idea, which is exactly the kind of result a benchmark paper should publish. We note the analysis is scoped to $n{=}612$ annotated rows and is therefore underpowered, so the confidence intervals are wide; we report it either way per pre-registration.

4 Reproducibility and Integrity

Because this field is under active scrutiny for inflated metrics, we built integrity guarantees into the benchmark protocol rather than treating them as an afterthought. Every result was pre-registered and hash-frozen before scoring: the primary benchmark carries a pre-registration SHA-256 fingerprint, and the recalibration and risk-control analyses each carry their own pre-registration hashes with success bands declared before any number was computed. Receipts are content-hashed, so any post-hoc alteration is detectable.

We commissioned an independent audit that re-derives the benchmark surface directly from the raw 16.87 GB single-cell data with fresh code that does not reuse the pipeline's cache. For the audited donor fold, the audit reproduces the effect-target multiset to within 10^{-6}, with a maximum sorted absolute difference of exactly 0.0 and matching row counts and feature columns, establishing that the cached benchmark surface is a faithful materialization of the raw data and not a fabrication. A second re-scoring audit confirms that the conformal arithmetic on that surface is reproducible.

Finally, in the interest of full disclosure, we retract two earlier surrogate results from this project's development history that did not survive scrutiny. Reporting a self-retraction alongside a tie and a negative result is uncomfortable, but it is the honest record of the measurement, and it is the standard we believe perturbation UQ should be held to.

5 Limitations

This benchmark has real limits that bound its claims. First, the donor-blocked design has only *four donors*, and the exchangeable unit for the donor-shift guarantee is the donor. This is adequate for measuring marginal calibration under LODO, which is our purpose, but it is underpowered for any inferential claim that requires more than a handful of exchangeable units, including the risk-control certificate in the LODO regime, which is honestly vacuous at $n{=}4$. Any extension that treats the donor as the inferential unit inherits this ceiling.

Second, the genetics-conditioning experiment is a negative result scoped to 612 annotated rows and is underpowered; its confidence intervals are wide, and we do not claim that no genetics-based conditioning could ever help, only that this natural instantiation did not on this data.

Third, and most importantly, *no biological outcome is validated in scope*. This is a calibration and selective-prediction benchmark. The $CD4^{+}$ T-cell and autoimmune context motivates the choice of data and donor structure, but we make no claim that any perturbation is a validated disease target. Calibrated uncertainty about a perturbation-effect estimate is not evidence about biological efficacy.

Fourth, the transfer study uses a single external assay (K562), so the recalibration protocol is demonstrated on one cross-assay pair rather than a broad panel; generalization across many assays remains future work.

6 Conclusion

We have presented a calibration and selective-prediction benchmark for perturbation-effect uncertainty that leads with an honest tie: distribution-free calibration methods are statistically indistinguishable from one another on a fair, point-prediction-controlled donor-blocked surface, and conformal prediction earns no measurable edge over quantile regression. The benchmark nonetheless resolves a clear and reproducible contrast, showing that parametric priors miscover under donor shift while distribution-free methods do not. Alongside the core head-to-head, we contribute a cross-assay transfer protocol with an honest failure mode and its fix, a distribution-free risk-control analysis that quantifies the price of exchangeability rather than hiding it, and a pre-registered negative showing that an external genetics taxonomy does not sharpen intervals. At a moment when the field is alarmed about inflated perturbation metrics [Ahlmann-Eltze et al., 2025, Wu et al., 2024], we hope a benchmark that foregrounds its tie, its negative, its exchangeability limits, and its self-retraction offers a useful and credible standard for measuring calibrated uncertainty in single-cell perturbation prediction.

Availability. All benchmark receipts are content-hashed and deterministically reproducible from the committed pipeline, and the external transfer analyses reproduce from cached group statistics for the K562 assay.

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